MIL-P-37361
4 February 1977

MILITARY SPECIFICATION

PLIERS, DENTAL, CONTOURING AND CRIMPING, No. 137

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a dental pliers suitable for contouring and crimping bands and crowns.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.

MILITARY

MIL-P-116 - Methods of Preservation-packaging.
MIL-L-10547- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th St. Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

FEDERAL

FED-STD-151 - Metal; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

20500 - PLIERS, DENTAL, NO. 137

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

E 92 - Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association apecifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General requirements. The pliers shall be in accordance with applicable requirements of GG-I-526 with the exceptions and with the additional requirements specified herein.

3.2 Material. Pliers shall be fabricated from either carbon steel or stainless steel as specified in table I.

3.3 Hardness. After proper stress relieving, the pliers shall have the Rockwell hardness as specified in table I for the applicable material. Hardness shall be determined as specified in 4.3.2.

3.4 Distortion. The handle of the pliers shall not show pliability or distortion under normal usage.

3.5 Style, design and dimensions. The pliers, designated as No. 137 Gorden dental pliers, shall be suitable for general contouring work in the dental laboratory. Pliers shall conform to Defense Personnel Support Center (DPSC) Drawing No. 20500.

3.5.1 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, function, fit, assembly, and operation of the item. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

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TABLE I. Material, hardness and plating.

Material		Rockwell Hardness	Plating
Carbon	Carbon content: 0.40 to 1.10 percent Manganese content: 0.60 to 0.90 percent except when carbon content is over 0.90 percent, the manganese content shall be a maximum of 0.50 percent. Phosphorous content: Maximum of 0.040 percent Sulfur content: Maximum of 0.050 percent	C40 to C45, with the exception of the screw or pin. When the carbon content is over 0.90 percent, the hardness shall be C48 to C52.	Plating shall be in accordance with any one of the following three methods: 1. Chromium or nickel plated in accordance with GG-I-526 except the minimum thickness of coating on the box-lock or lap joint area shall be as follows: Copper plus nickel--0.0007 inch Final nickel--------0.0003 inch Chromium----------0.000009 inch 2. Nickel plated with a minimum thickness of 0.001 inch and maximum thickness of 0.003 inch. 3. Nickel plated with a minimum thickness of 0.0004 inch and chromium overplated with a minimum thickness of 0.00001 inch.
Stainless steel	Class 4, Type 420	C40 to C48, with the exception of the screw or pin.	None
	Class 4, Type 410	C40 to C45, with the exception of the screw or pin.	None

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3.6 Locks. The joint of the pliers shall be box-lock or lap-joint type. The edges of joints shall be properly beveled to prevent pinching of or injury to tissue.

3.7 Finish. In addition to the requirements of GG-I-526, the pliers shall have a mirror finish except for the handles which shall have a mirror or satin finish. The inside surfaces of the jaws of the pliers shall be smoothly contoured as specified in the drawing.

3.8 Plating. Plating shall be in accordance with table I, as applicable for the material specified.

3.9 Corrosion resistance. The pliers shall meet the boil test requirements of GG-I-526 when tested as specified in 4.3.3.1. Pliers fabricated from stainless steel shall also meet the copper sulfate test requirements of GG-I-526 when tested as specified in 4.3.3.2.

3.10 Identification marking. The pliers shall be marked with the number "137". Marking shall be permanent and legible. Stainless steel pliers of foreign manufacture shall also be marked with the word "stainless".

3.11 Workmanship. Workmanship shall be first class throughout. The pliers shall be free from defects which may detract from their appearance or impair their serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection rquirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless diapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metal, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component or material, the date of testing, the test method, individual test results, the specification requirements, the lot number, the lot size and the sample size.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Pliers shall be examined for defects including, but not limited to, those listed in table II.

TABLE II. Classification of defects.

Categories	Defects
Major A:	
101	Style and design not as specified.
102	Finish not as specified.
103	Plating of plated pliers not free from flaking or blisters.
104	Plating of plated pliers incomplete, porous, or not continuous.
105	Jaws do not align properly.
106	Joint of pliers not free from stiffness.
107	Joint of pliers has excessive play.
108	Edges of the joint of pliers sharp (not properly beveled).
109	Pliers not free from cracks.

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TABLE II. Classification of defects. (cont'd)

Categories	Defects
Major B:	
151	Dimensions not as specified.
Minor:	
201	Identification marking not present, legible, or complete.
202	Pliers not permanently marked with the name or trademark of the contractor.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table III. The unit of product for examination shall be one pliers.

TABLE III. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor	II	2.5

4.2.2 Testing. Pliers shall be tested as specified in table IV and 4.3. Inspection levels and acceptable quality levels (AQL's) shall be as listed in table IV. The unit of product for testing shall be one pliers.

TABLE IV. Test criteria.

Characteristic	Requirement	Test procedures	Inspection level	AQL
Material	3.2	4.3.1	**	*
Hardness	3.3	4.3.2	S-2	1.5
Boil Test	3.9	4.3.3.1	S-2	1.0
Copper sulfate	3.9	4.3.3.2	S-2	1.0

* Acceptance number shall be zero for all sample sizes.

**A sufficient amount from each lot of metal used in the construction of the pliers shall be selected for chemical analysis.

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4.2.3 Packaging inspection. The inspection of preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisons of the applicable container specification and marking requirements of MIL-STD-129.

4.3 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

4.3.1 Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

4.3.2 Hardness. Hardness shall be determined in accordance with ASTM E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.3.3 Corrosion resistance.

4.3.3.1 Boil test. The pliers shall be subjected to the boil test, as specified in GG-I-526.

4.3.3.2 Copper sulfate test. The pliers shall be subjected to the copper sulfate test, as specified in GG-I-526.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Pliers shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each pliers shall be wrapped in neutral tissue paper or other suitable anticorrosive materials. Pliers shall be further packaged in accordance with MIL-P-116, method IC-2.

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5.1.1.4 Intermediate packaging. Six (6) pliers shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. One hundred forty four (144) pliers, shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. One hundred forty four (144) pliers, shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-543-3750	PLIERS, DENTAL, No. 137, 5-1/4 inch

6.3 Supersession data. This specification includes the requirements of Purchase Description No. 10, dated 14 August 1968.

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA - DM

Project No. 6520-1414

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/2083

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-37361 PLIERS, DENTAL, CONTOURING AND CRIMPING, NO. 137

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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